

An Internship Report

Of

Python Programming

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (CSE)



By

Roll Number of Student:- 25SCS1003003488

Name of Student: Kartik Raj Singh

Section :- 2CSE8

Candidate's Declaration

I, Kartik Raj Singh, do hereby declare that the internship report titled “Python Programming” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: Kartik Raj Singh

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Acknowledgement

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank CodeAlpha for providing me the opportunity to undergo this virtual internship in Python Programming, and for the guidance and support extended to me throughout the internship period. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: Kartik Raj Singh

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3. INTERNSHIP COMPLETION CERTIFICATE

 CODE — ALPHA — YOUR DREAM OUR PASSION	 MINISTRY OF CORPORATE AFFAIRS GOVERNMENT OF INDIA
CERTIFICATE OF COMPLETION	
This Certificate Is Proudly Presented To Kartik Raj Singh	
This Certificate recognizes the successful completion of the CodeAlpha Virtual Internship Program in Python Programming, demonstrating dedication and sincerity throughout the program. We appreciate the valuable efforts and wish continued success in all future endeavors. 1st August 2026 to 30th August 2026	
 CEO & Co-Founder	19th September 2026 Date of Issue
 SCAN TO VERIFY THIS CERTIFICATE Student ID: CA/DFI/215909	

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the internship undertaken as a Python Programming Intern at CodeAlpha, a software development company that runs virtual internship programs across domains such as web development, app development, machine learning, and Python programming. The internship was carried out remotely from 1 August 2026 to 30 August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of the internship was to strengthen practical, hands-on Python programming skills by building small, self-contained projects covering core concepts such as control flow, functions, file handling, and automation. On successful completion of the assigned tasks, an Internship Completion Certificate was issued by CodeAlpha.

4.2 Organization Profile

CodeAlpha is a software development company that offers virtual internship programs designed to give students hands-on, real-world programming experience. The internship structure typically provides a list of tasks within a chosen domain, of which interns are required to complete a minimum number within the internship period, upload their source code to GitHub, and share their progress on LinkedIn. In addition to the completion certificate, CodeAlpha internships include perks such as a letter of recommendation, resume-building guidance, and job placement support.

4.3 Problem Statement

New programmers often understand language syntax without having applied it to complete, working programs. A common gap is the absence of guided, real, hands-on tasks that require combining multiple concepts, such as loops, conditionals, functions, and file handling, into a single working solution. This internship addressed that gap by assigning a set of small but complete Python projects, each requiring the practical application of specific programming concepts from start to finish.

4.4 Project Objectives

- To strengthen core Python fundamentals through hands-on project work.
- To build a complete, playable console-based game applying loops, conditionals, and string handling.
- To automate a real, repetitive file-management task using Python's `os` and `shutil` modules.
- To design and implement a simple rule-based chatbot using functions and conditional logic.
- To practice structuring, documenting, and version-controlling Python projects using Git and GitHub.
- To apply the above skills on real, assigned tasks during the internship period.

4.5 Scope of the Project

The scope of this internship covers three self-contained Python programs, each addressing a different core programming concept: an interactive console game, a file-automation script, and a rule-based conversational program. Each task was developed independently, tested locally, and uploaded to a dedicated GitHub repository, with the internship not extending to any client-facing or production system.

4.6 Technologies and Tools Used

- Programming Language: Python 3
- Core Modules: random, os, shutil
- Concepts Applied: loops, if-else/if-elif conditionals, functions, lists, strings, file handling
- Development Environment: VS Code
- Version Control: Git and GitHub
- Documentation: Markdown (README files)

4.7 Task Breakdown

The internship required completing a minimum of two to three tasks from a provided Python programming task list. The following three tasks were completed:

Task	Title	Description
1	Hangman Game	A text-based word-guessing game where the player guesses a word one letter at a time, with a limit of 6 incorrect guesses, built using random word selection, a while loop, and string tracking.
2	Task Automation with Python Scripts	A script that automates moving all .jpg files from a user-specified source folder to a user-specified destination folder, using the os and shutil modules.
3	Basic Chatbot	A simple rule-based chatbot that responds to greetings and a few predefined questions using if-elif logic wrapped inside a function, running in a continuous loop until the user says 'bye'.

4.8 Methodology

The internship followed a structured methodology combining orientation, hands-on task execution, and periodic review, as summarized below.

Phase	Activity	Description
1	Orientation	Reviewing the task list provided by CodeAlpha and understanding submission requirements.
2	Task Selection	Selecting the Hangman Game, Task Automation, and Chatbot tasks from the given list.
3	Development	Writing, testing, and debugging each Python script locally.
4	Documentation	Preparing a README for each task explaining its goal, concepts used, and how to run it.
5	Version Control	Uploading each completed task to a GitHub repository as a separate folder.
6	Finalization	Reviewing all tasks and the repository structure before internship completion.

4.9 Expected Outcomes

- Practical, working knowledge of core Python programming concepts.
- Ability to independently plan, build, and test small console-based applications.
- Hands-on experience automating a real file-management task.
- Improved proficiency in structuring and documenting projects with Git and GitHub.
- A completed internship and certificate demonstrating readiness for further programming coursework and roles.

4.10 Certificate of Completion and Communication Proof

The Internship Completion Certificate issued by CodeAlpha, confirming successful completion of the Virtual Internship Program in Python Programming from 1 August 2026 to 30 August 2026, is shown below.



5. Bibliography/References

1. CodeAlpha, Certificate of Completion, Kartik Raj Singh, Python Programming Virtual Internship, issued 19 September 2026.
2. Python Software Foundation, Python 3 Documentation, docs.python.org.
3. GitHub Docs, github.com/docs.